Hyperspectral Image Classification Using Groupwise Separable Convolutional Vision Transformer Network

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***Abstract*— Recently, vision transformer (ViT)-based deep learning (DL) models have achieved remarkable performance gains in hyperspectral image classification (HSIC) due to their abilities to model long-range dependencies and extract global spatial features. However, ViT is built with a stack of Transformer blocks and faces the challenge of learning a large number of parameters when processing hyperspectral data. Besides, the inherent modeling of global correlation in Transformer ignores the effective representation of local spatial and spectral features. To address these issues, we propose a lightweight ViT network known as groupwise separable convolutional ViT (GSC-ViT). First, a groupwise separable convolution (GSC) module, which is a combination of grouped pointwise convolution (GPWC) and group convolution, is designed to significantly decrease the number of convolutional kernel parameters, and effectively capture local spectral–spatial information in hyperspectral image (HSI). Second, a groupwise separable multihead self-attention (GSSA) module is employed to substitute the conventional multihead self-attention (MSA) in ViT, in which the groupwise self-attention (GSA) provides local spatial feature extraction, and the pointwise self-attention (PWSA) provides global spatial feature extraction. Third, a simple pointwise layer with enhanced skip connection mechanism is employed to substitute the multilayer perceptron (MLP) layer in all Transformer blocks of ViT, so as to eliminate unnecessary nonlinear transformations and facilitate the fusion of features derived from GSC and GSSA modules. Extensive experiments on four benchmark hyperspectral datasets reveal that our GSC-ViT can achieve surprising classification performance with relatively few training samples as compared with some existing HSIC approaches. The source code is available at https://github.com/flyzzie/TGRS-GSC-VIT.**

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***Index Terms*— Groupwise separable convolution (GSC), groupwise separable multihead self-attention (GSSA), hyperspectral image classification (HSIC), vision transformer (ViT).**

I. Introduction

HYPERSPECTRAL image classification (HSIC) constitutes a significant task in the realm of remote sensing [1], its primary objective is to classify and identify ground objects using hyperspectral image (HSI). Nowadays, HSIC offers crucial technical support for environmental monitoring [2], [3], urban planning [4], [5], agricultural production [6], [7], [8], and national defense [9], among other domains [10].

Over the past few decades, researchers have come up with a variety of HSIC methods. According to the feature extraction modes, these methods generally fall into two categories: handcrafted methods and deep learning (DL)-based methods.

Traditional handcrafted methods encompass statistical classification [11], decision tree [12], support vector machine (SVM) [13], K-nearest neighbor (KNN) [14], Gabor filter [15], and random field [16], etc. Although they exhibit benefits of well-established algorithms, ease of understanding, and intuitive implementation, their limitation lies in the insufficient in dealing with the inherent nonlinearity of hyperspectral data.

DL-based methods encompass convolutional neural network (CNN) [17], [18], and vision transformer (ViT). Among, Hu et al. [19] proposed a spectral-domain CNN for HSIC. Lee and Kwon [20] proposed a 2-D CNN to jointly exploit spatial and spectral features of HSI. Due to the deep fully convolutional structure, the number of parameters and the computational complexity are high. Furthermore, Hamida et al. [21] employed 3-D CNN to synchronously process spectral and spatial information. Nevertheless, 3-D convolution operation augments computational load of the model. Roy et al. [22] amalgamated 2-D and 3-D CNNs to build a HybridSN model, demonstrating favorable overall performance. The aforementioned CNN-based models excel in extracting local spectral–spatial information from HSI. However, they also exhibit intricate structures, substantial model parameters, and high computational demands. Typically, a large number of training samples are required to ensure optimal learning performance; otherwise, these models

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are prone to overfitting, especially when the training data are limited.

Generally, shallow-level features are inclined to characterize primitive feature elements, such as points, lines, and edge contours. These feature elements implied clues to spatial information in HSI. On the other hand, deep-level features tend to emphasize the semantic information of an image, while less attention is paid to the details of spatial structures. At this point, overly complex or deeper network models may not be the best choice for HSIC tasks.

Recently, to ease the network complexity, some lightweight operations have been carried out on traditional CNN models, and part of lightweight models have been successfully applied to hyperspectral classification. Among, SqueezeNet [23] employed pointwise (1×1) convolutions initially to reduce the dimensionality of feature maps. Subsequently, convolutions with different scales (1×1 and 3×3) were applied to the dimensionally reduced feature maps, resulting in the significant compression of the number of model parameters, approximately $1/50$ of the AlexNet [24]. Then, Fang et al. [25] combined the squeeze convolution module with multibias module to further reduce the model complexity. Moreover, MobileNet [26] introduced depthwise separable convolution (DSC), where DSC combined depthwise and pointwise convolutions (PWCs). Gao et al. [27] incorporated multiscale convolutional kernels into a single DSC to extract rich features by linking shallow and deep features through residual connections. According to the characteristics of hyperspectral data, Zhao et al. [28] proposed the concept of grouped separable convolution, which involves the integration of grouped convolution (GC) with PWC. Here, GC partitions hyperspectral data into nonoverlapping groups according to spectral dimension, and subjects each group to individual standard convolution operation, which accentuates the feature representation between adjacent spectral bands within each group while concurrently reducing the overall parameters quantity.

The above lightweight strategies relieve the model complexity and parameter quantity to some extent. However, CNNs are sensitive to local information but pool at extracting global information due to their inherent local receptive fields. In HSIC, training samples typically have a cube style. Given that the size of the receptive field is less than the cube size, it will lose global sensing of the entire cube. To address this issue, ViT [29] adopts a multihead self-attention (MSA) [30] mechanism to address global dependencies in the image, showcasing remarkable performance in image classification tasks [31]. Moreover, ViT exhibits excellent scalability, effortlessly accommodating various image sizes. Nevertheless, due to its substantial model architecture, it typically necessitates pretraining on large-scale image datasets, followed by fine-tuning for specific tasks.

To address the training challenges encountered by ViT, a number of improvement strategies have been introduced. Among, DeiT [32] employs a teacher–student training strategy for the distillation of pure transformer. Swin Transformer [33], [34] introduces a hierarchical convolution-ViT structure to address multiscale information, and incorporates a shift window to the self-attention mechanism to enhance the extraction of local information with low computational overhead. Furthermore, MobileViT [35] integrates convolution and ViT, and employs a "unfold-transformer-fold" strategy to reduce computational overhead of self-attention mechanism. SpectralFormer [36], as the first transformer model for HSIC, involves a groupwise spectral embedding (GSE) module to acquire localized spectral representations, and a cross-layer adaptive fusion (CAF) module to combine features of shallow and deep layers. However, SpectralFormer still exhibits a notable performance gap compared to the most advanced methods. After, Sun et al. [37] introduced a fusion architecture of CNN and transformer for HSIC, in which CNN modules are leveraged to extract shallow-level spectral–spatial features, and transformer modules focus on extracting deep-level semantics that align with the sample distribution. Mei et al. [38] refined the convolution-ViT composite model and restricted ViT to a local spectral–spatial context by using GC. However, due to the continuous use of MSA in the stacking of transformer blocks, the model maintains a substantial parameter quantity and computational demands. Recently, Roy et al. [39] introduced a novel model called MorphFormer, which combines spectral and spatial morphological convolution with transformer, thus enhance the interaction between image tokens and class tokens on structure and shape information in the HSI.

The local feature processing capability of CNNs and the global receptive field of ViT represent distinct advantages in their respective models. The combination of CNN and ViT yields a model with robust feature extraction capabilities as well as a recognition acknowledged by prior researchers. Building upon the architecture integrating CNN and ViT, we propose a novel lightweight convolution-ViT network model referred to as "Groupwise Separable Convolutional-ViT" (hereafter dubbed "GSC-ViT"). In a nutshell, our work encompasses three main innovative contributions as follows.

1) First, we constructed a lightweight convolution-ViT network and introduced a groupwise separable convolution (GSC) module to substitute traditional convolution. Here, GSC combines grouped PWC (GPWC) and GC, thus not only highlight the local spectral–spatial joint features of adjacent bands within each group but also significantly reduce the number of convolutional kernel parameters.
2) Second, aiming at the ViT blocks, we designed a groupwise separable multihead self-attention (GSSA) module, which disassembles the conventional MSA module into two more lightweight components: groupwise self-attention (GSA) and pointwise self-attention (PWSA). GSA is employed to extract local spatial feature, while PWSA captures global spatial information.
3) Third, we introduced a simple PWC layer with an enhanced residual connection to substitute the multilayer perceptron (MLP) layer in transformer blocks of ViT. This substitution eliminates unnecessary nonlinear transformations and facilitates the fusion of features from GSC and GSSA modules.

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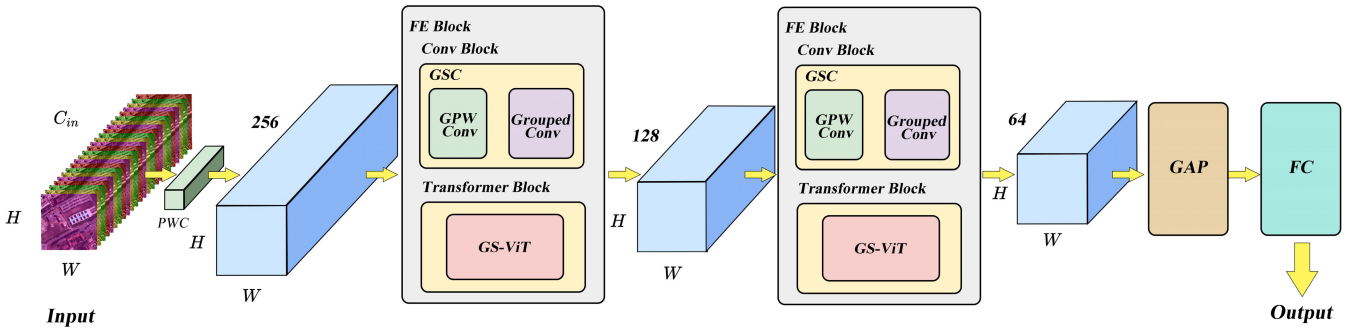


Fig. 1. Schematic of the proposed GSC-ViT model, in which C_{in} denotes the number of input channels, "PWC" refers to the pointwise convolution and "GPW Conv" refers to the grouped pointwise convolution, "Grouped Conv" stands for the grouped convolution, "GS-ViT" denotes the grouped separable ViT, "GAP" block denotes the global average pooling layer, and "FC" refers to the fully connected layer.

Extensive experiments are conducted on four authentic hyperspectral datasets. The experimental outcomes demonstrate that GSC-ViT model exhibits minimal training costs. Compared to several state-of-the-art CNN and ViT models previously applied in HSIC, our novel model achieves competitive classification accuracy with fewer training samples required.

The remainder of this article is organized as follows. Section II presents the architecture of the proposed GSC-ViT model in a step-by-step fashion. Section III presents a comprehensive experimental assessment of the GSC-ViT model. Section IV concludes this article with some remarks and hints at plausible future research lines.

II. Proposed GSC-ViT Model

This section provides a detailed exposition of our proposed GSC-ViT model. First, we elucidate the overall framework of the GSC-ViT model and its workflow for HSIC. Second, we delve into the motivation of several key components in the GSC-ViT architecture.

A. Framework of GSC-ViT Model

The overall architecture of the GSC-ViT model is delineated in Fig. 1, which can be broken down into four key components: the first is a spectral calibration (SC) block, the second comprises two feature extraction blocks, the third is a global average pooling (GAP) block, and the final component is a SoftMax classifier reliant on fully connected (FC) layers.

In GSC-ViT, the input is manifested as a 3-D cube, whose spectral dimension (also called channel dimension) size varies depending on the dataset, so we adopt PWC to calibrate the spectral dimension to a fixed value for subsequent channel grouping. The second component consists of two feature extraction blocks, a lightweight GSC block and a lightweight groupwise separable ViT (GS-ViT) block. The GSC block is composed of GPWC and GC modules. Its role is to conduct local feature extraction, joint spectral–spatial feature extraction, and channel reduction. When the feature map traverses each convolution layer, the number of channels systematically reduces by half. The GS-ViT block is deployed after the GSC block, with the help of the GSSA model, GS-ViT is responsible for extracting the local–global spatial information and integrating channel information. In the GAP layer, the GAP operation is applied to shrink the output feature map to a spatial size of 1×1. Lastly, the feature map is transformed into 1-D vector, and fed to an FC layer-based classifier for the final output of classification results.

B. Spectral Calibration Block

The purpose of the SC block is to transform the number of input channels to a fixed value (as different HSI data may have different number of channels). Here, we use PWC to implement SC block. PWC is a convolution operation with a spatial size of 1×1, so it does not alter the spatial dimensions of input feature maps during the convolution process. As shown in Fig. 3, PWC transforms the input feature map from a size of $C_{\text{in}} \times H \times W$ to $C_{\text{out}} \times H \times W$, in this case, C_{out} number of convolutional kernels with size of $C_{\text{in}} \times 1 \times 1$ are used.

Two reasons inspired us to employ PWC for channel calibration. First, PWC has a minimal number of parameters for channel expansion. For the above example, the parameters of convolutional kernel involved amount to $C_{\text{in}} \times C_{\text{out}}$. Comparatively speaking, standard convolution with spatial dimension larger than 1 or FC layer for channel calibration would significantly increase the number of parameters. Second, PWC ensures that the spatial dimensions are unchanged while only the spectral dimensional features are integrated, thus facilitating subsequent spectral-spatial feature extraction.

C. Convolution Block in the Feature Extraction Block

In each FE block, a GSC block is first performed, which consists of two main components: GPWC and GC.

1) Grouped Convolution: Compared with standard convolution, GC has different convolution operations. As illustrated in Fig. 4, the GC first divides the input feature map into g nonoverlapped and equal-size groups along the channel dimension. Then, a standard convolution operation is applied independently to each group. More specifically, for an input feature map $\mathbf{F}$ with shape of $C_{\text{in}} \times H \times W$, $\mathbf{F}$ is first split into g nonoverlapped groups along the channel dimension

$$\mathbf{F} = \{\mathbf{F}_1, \ldots, \mathbf{F}_i, \ldots, \mathbf{F}_g\} \tag{1}$$

where g is the number of groups, and each grouped feature map $\mathbf{F}_i$ has a size of $(C_{\text{in}}/g) \times H \times W$. For each $\mathbf{F}_i$, standard

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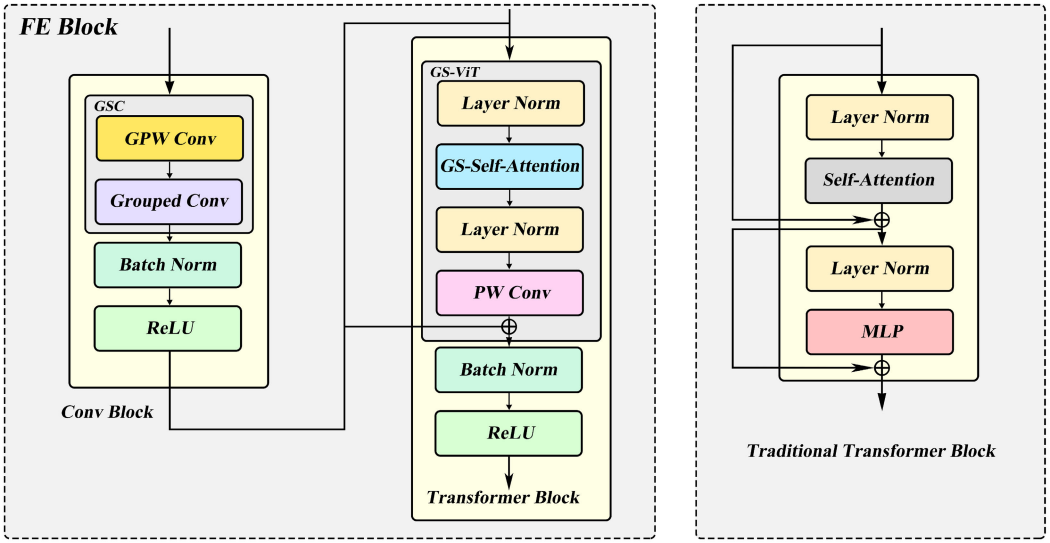


Fig. 2. Structure of feature extraction block is on the left side of the vertical dotted line. The components of the traditional transformer block is on the right side of the dotted line for comparison. "Batch Norm" refers to the batch normalization, "ReLU" stands for the activation function, "Layer Norm" represents the layer normalization, "MLP" refers to the multilayer perceptron.

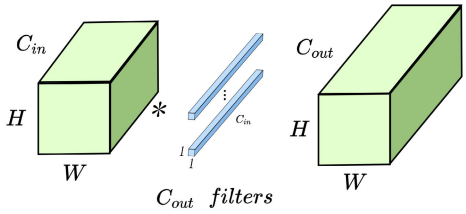


Fig. 3. Convolution process of PWC.

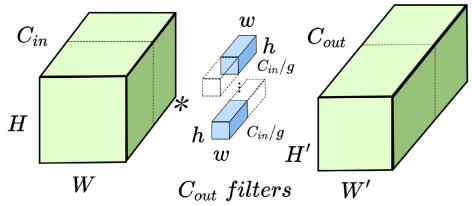


Fig. 4. Convolution process of GC.

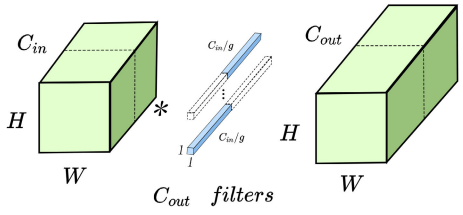


Fig. 5. Convolution process of GPWC.

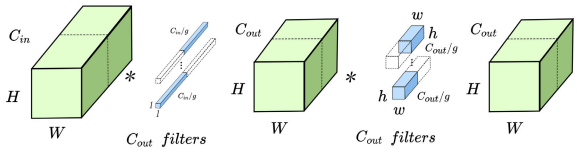


Fig. 6. Convolution process of GSC.

convolutions are applied individually, resulting in the output of each group. Subsequently, the output features from all groups are concatenated in sequential order, denoted as $\mathbf{F}'$

$$\mathbf{F}' = \text{Concat}\left(\mathbf{F}'_1, \ldots, \mathbf{F}'_i, \ldots, \mathbf{F}'_g\right). \tag{2}$$

In the standard convolution, given the input feature map has a size of $C_{\text{in}} \times H \times W$, the corresponding output feature map has a size of $C_{\text{out}} \times H' \times W'$, so it requires C_{out} standard convolution kernels with size of $C_{\text{in}} \times h \times w$. In contrast, GC maintains the same input and output settings. It uses the same C_{out} number of standard convolution kernels, but with a smaller size of $(C_{\text{in}}/g) \times h \times w$ and divided into g groups. As a result, GC reduces the total number of convolution kernel parameters from $C_{\text{in}} \times C_{\text{out}} \times h \times w$ to $(C_{\text{in}}/g) \times C_{\text{out}} \times h \times w$. Essentially, the parameter quantity of GC is only $1/g$ of that in standard convolution.

2) Grouped Pointwise Convolution: GPWC is a specific version of GC, which applies GC by using a 1×1 spatial kernel. The key difference lies in its ability to preserve the spatial dimensions during the convolution process. As illustrated in Fig. 5, the number of parameters involved in the convolution kernels of GPWC is also reduced to $1/g$ of the PWC.

3) Combination of GPWC and GC to Form GSC: As shown in Fig. 6, in the convolutional block, the feature map initially undergoes the GPWC block, and then proceeds to the GC block by using 3×3 spatial kernels, thus forming a complete GSC block. At this point, some obvious benefits emerge: First, the convolutional kernel parameters involved in GSC are significantly reduced compared to standard convolution or GC. Assuming an input feature map has a size of $C_{\text{in}} \times H \times W$ and aiming for an output size of $C_{\text{out}} \times H \times W$ ($C_{\text{in}} > C_{\text{out}}$), the number of parameters for standard convolution would be $C_{\text{in}} \times C_{\text{out}} \times h \times w$. For GC with a group number of g, the number of parameters would be $C_{\text{in}}/g \times C_{\text{out}} \times h \times w$. However, when employing the combination of GSC blocks with a group number of g, the number of parameters becomes $C_{\text{in}}/g \times C_{\text{out}} \times 1 \times 1 + C_{\text{out}}/g \times C_{\text{out}} \times h \times w$. It is evident that GSC requires fewer parameters, and the ratio of parameter

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quantity between GSC and GC is shown in the following:

$$\frac{P_{\mathrm{GSC}}}{P_{\mathrm{GC}}}=\frac{\frac{C_{\mathrm{in}}}{g}\cdot C_{\mathrm{out}}+\frac{C_{\mathrm{out}}}{g}\cdot C_{\mathrm{out}}\cdot hw}{\frac{C_{\mathrm{in}}}{g}\cdot C_{\mathrm{out}}\cdot hw}=\frac{1}{hw}+\frac{C_{\mathrm{out}}}{C_{\mathrm{in}}} \quad (3)$$

where P_{GSC} is the parameters quantity of GSC, and P_{GC} refers to the parameters quantity of GC. As can be observed from (3), the ratio is related to the spatial dimensions of convolutional kernels in the GC block, as well as the ratio of the number of input and output channels. Considering the used convolutional kernels have a size of 3×3 and $(C_{\mathrm{out}}/C_{\mathrm{in}})$ is $(1/2)$, there has been $7/18$ reduction as compared to GC.

Second, GSC can simultaneously extract local spectral–spatial feature from HSIs. In the GSC block, the grouped size for GPWC and GC are consistent. GPWC is based on PWC, which partitions the channel dimension into nonoverlapping groups and performs independent convolution operations in each group. As a result, local channel feature can be extracted without altering relative spatial feature. On the basis of GPWC, the GC block further extracts local spatial feature without changing the channel number of feature map.

D. GS-ViT Block in the Feature Extraction Block

Traditional ViT suffers from high computational complexity and lacks local perception. Here, we propose a groupwise separable self-attention (GSSA), which decomposes the standard MSA into two parts: GSA and PWSA. This decomposition greatly reduces the computational burden of self-attention operation on token and emphasizes the effective extraction of global and local spatial features. Additionally, considering that HSIC does not rely strongly on highly abstract semantic information, we substitute the MLP layers with PWC layers, and enhance the skip connection scheme in the transformer block. With these modifications, we can eliminate unnecessary nonlinear transformations, and apply only a single transformer block for feature learning, resulting in further reduction of the overall model parameters quantity.

1) Standard MSA: Assuming the input feature matrix is denoted as $A_0\in\mathbb{R}^{H\times W\times C}$, where C represents the number of channels, H and W represent the spatial scale. First, we squeeze the spatial dimension to form a 2-D matrix $A\in\mathbb{R}^{HW\times C}$. Then a linear transformation is performed on A to generate three distinct matrices: the query matrix Q, the key matrix K, and the value matrix V, they all have the same size of $\mathbb{R}^{HW\times C}$. The specific transformation formulas are shown in the following:

$$Q=A\times W_Q,\ K=A\times W_K,\ V=A\times W_V \quad (4)$$

where $W_Q, W_K, W_V\in\mathbb{R}^{C\times C}$. Next, self-attention is performed for each of the matrices Q, K, V, as specified by the following:

$$\mathrm{SA}=\mathrm{Attention}(Q,K,V)=\mathrm{SoftMax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (5)$$

where d_k stands for the dimension of matric K, and $\mathrm{SA}\in\mathbb{R}^{HW\times C}$.

MSA involves mapping multiple sets of weight matrices for Q, K, and V, using the same operation to calculate multihead attention values. Then, the results of each head of attention are concatenated together, as illustrated in the following:

$$\mathrm{MSA}(Q,K,V)=\mathrm{Concat}(\mathrm{SA}_1,\ldots,\mathrm{SA}_i,\ldots,\mathrm{SA}_n)W. \quad (6)$$

In this context, each head is evenly divided along the channel dimension. Then perform the multihead division even before the SA operation. Here, $\mathrm{SA}_i\in\mathbb{R}^{HW\times C/n}$, with n represents the number of heads, and $W\in\mathbb{R}^{C\times C}$.

We conduct computational complexity analysis on the aforementioned MSA process. For a matrix linear transformation $X\times B=Y$ with $X\in\mathbb{R}^{M\times N}$, $B\in\mathbb{R}^{N\times L}$, and $Y\in\mathbb{R}^{M\times L}$, the computational complexity is proportional to $M\times N\times L$. Therefore, in (4), the computational complexity is $3\mathrm{HWC}^2$, in (5), the complexity of $\mathrm{QK}^T(Q\in\mathbb{R}^{\mathrm{HW}\times C}, K^T\in\mathbb{R}^{C\times\mathrm{HW}})$ is $(\mathrm{HW})^2C$ and the complexity of $\mathrm{EV}(E=\mathrm{SoftMax}((\mathrm{QK}^T/(d_k)^{1/2}))\in\mathbb{R}^{\mathrm{HW}\times\mathrm{HW}}, V\in\mathbb{R}^{\mathrm{HW}\times C})$ is also $(\mathrm{HW})^2C$, in (6), the complexity is HWC^2. Overall, the computational complexity of standard MSA can be noted as follows:

$$\mathrm{Complex}(\mathrm{MSA})=4\mathrm{HWC}^2+2(\mathrm{HW})^2C. \quad (7)$$

2) Groupwise Self-Attention: Different from performing MSA simultaneously on all tokens within a patch, in GSA of GSSA, to extract local information while reducing computational complexity, we first group patches along the spatial dimension. As illustrated in Fig. 7, the spatial dimensions of the input feature map are first partitioned into g equally sized submaps, each submap refers to as a group. Then, for each group, a "group token" (a singular pixel that serves as the core feature representation of the corresponding group) is automatically generated. Lastly, MSA is performed on all feature tokens within each group, including both the spatial pixel tokens and the group token. Importantly, the MSA is executed independently for each group. The concept of "group token" is introduced here to achieve a global representation of the features within each group. The initial value of this group token can be either random or zero. In the process of GSA, each group token can undergo self-attention with all tokens within the group, resulting in a representation of global information. The essence of GSA is the extraction of hyperspectral spatial local features. The use of group tokens is aimed at assigning global weights to each feature submap during the process of PWSA. Therefore, the GSSA process represents a spatial feature extraction that combines local and global considerations. The formal expression for implementing GSA is shown in the following:

$$\mathrm{GSA}(j)=\mathrm{MSA}\left(Q_j,K_j,V_j\right) \quad (8)$$

where j represents the jth group in GSA. Assuming the input size of feature map is $C\times H\times W$, with each group having spatial dimensions of $M\times M$, the total number of groups is $N=(\mathrm{HW}/M^2)$ and $Q_j, K_j, V_j\in\mathbb{R}^{(M^2+1)\times C}$. Following (7), the computational complexity for each group is shown in the following:

$$\mathrm{Complex}_j(\mathrm{GSA})=4\left(M^2+1\right)C^2+2\left(M^2+1\right)^2C. \quad (9)$$

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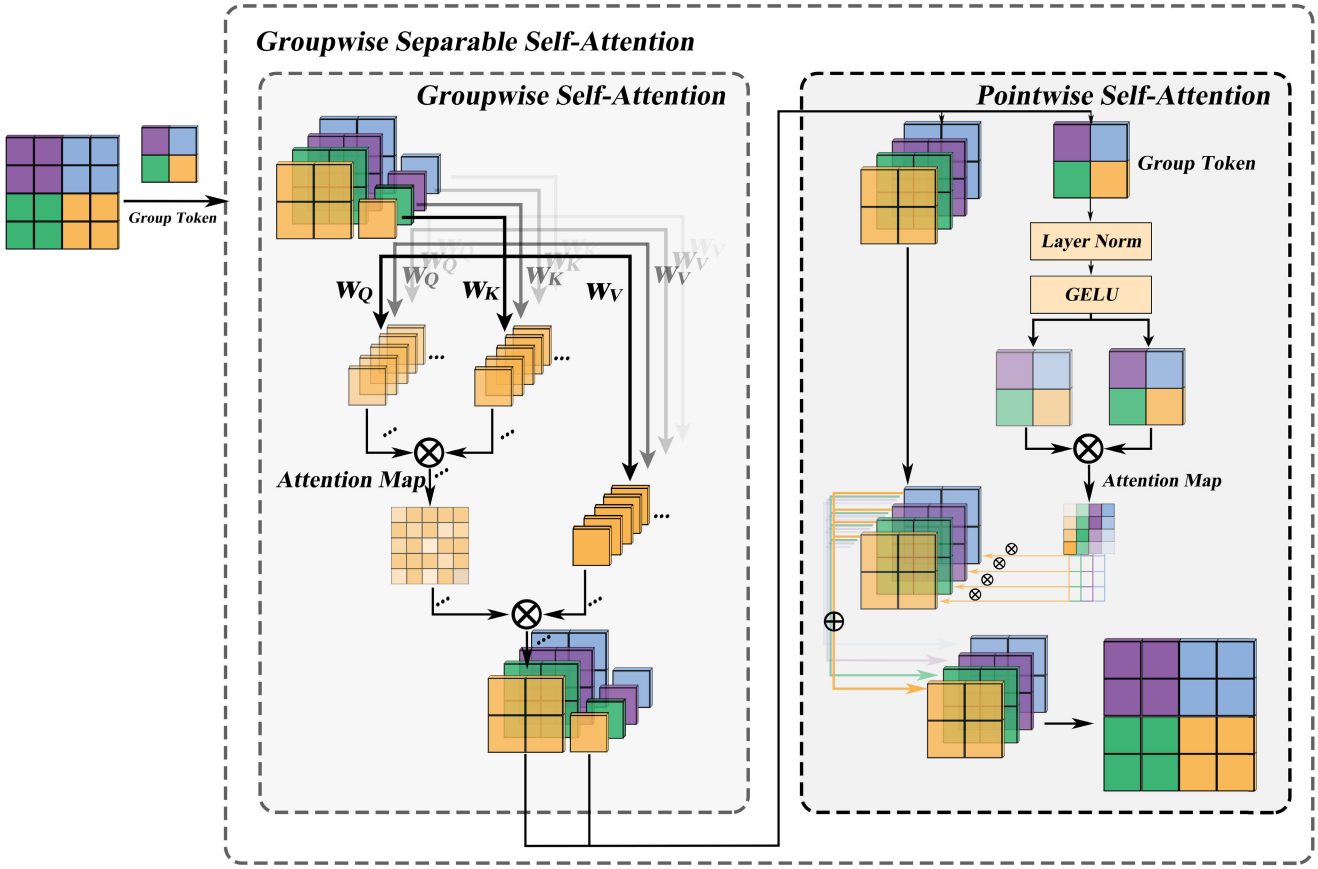


Fig. 7. Schematic of the GSSA in the GS-ViT block.

Therefore, in the GSA stage, the total computational complexity should be the product of the computational complexity for each group and the number of groups, as denoted by $\text{Complex(GSA)} = N \cdot \text{Complex}_j\text{(GSA)}$.

3) Pointwise Self-Attention: As illustrated in Fig. 7, PWSA is employed to build cross-group connections using group tokens. To be more specific, in the GSA stage, we acquire output group tokens, and then employ these group tokens to compute attention relationships between groups. After passing through the GELU activation function and Layer Normalization layer, attention weight matrices are generated. Attention calculations are then performed between the attention weight matrix and subfeature maps, which facilitates the fusion of global information. The PWSA process is described in the following:

$$\text{PWSA}\left(Q_{\text{gt}}, K_{\text{gt}}, z\right) = \text{MSA}\left(Q_{\text{gt}}, K_{\text{gt}}, z\right) \tag{10}$$

where $Q_{\text{gt}} = G(\text{gt})W_{Q_{\text{gt}}} = \text{LN}(\text{GELU}(\text{gt}))W_{Q_{\text{gt}}}$ and $K_{\text{gt}} = K(\text{gt})W_{K_{\text{gt}}} = LN(\text{GELU}(\text{gt}))W_{K_{\text{gt}}}$. Here $\text{gt} \in \mathbb{R}^{N\times C}$ represents the group token and $W_{Q_{\text{gt}}}, W_{K_{\text{gt}}} \in \mathbb{R}^{C\times C}$. LN stands for the Layer Normalization layer and GELU efficiently incorporates inactive information, which is defined as $\text{GELU} = x \cdot (1/2)[1 + \int_0^{(x/(2)^{1/2})} e^{-t^2}\,dt]$. Here, *attention* also functions as a standard self-attention operator, but it operates across the entire feature map in units of sub-feature maps (denoted as z).

After GSA, the computational complexity of PWSA for generating matrices Q and K is 2NC^2, the attention computation between Q and K is N^2C, and the computational complexity for the computation with z is $NHWC$. Overall, the computational complexity of PWSA can be described in the following:

$$\text{Complex(PWSA)} = 2\text{NC}^2 + N^2C + NHWC. \tag{11}$$

4) Complexity Analysis of GSSA: In the context of GSA, it is generally the case that the spatial dimensionality of the feature maps within each group (denoted as M) is not significantly small. Consequently, M^2 is typically much larger than the number of each group's token 1. The impact of the 1 on the overall computation can be negligible. Therefore, for convenience of comparison, it is acceptable to set $M^2 + 1 \approx M^2$ in (9). Therefore, after substituting $N = (\text{HW}/M^2)$ into the $\text{Complex(GSA)} = N \cdot \text{Complex}_j\text{(GSA)}$ and (9), we obtain $\text{Complex(GSA)} \approx 4\text{HWC}^2 + 2\text{HWM}^2C$. Adding the Complex(GSA) to Complex(PWSA) results in the following:

$$\begin{aligned}\text{Complex(GSSA)} &= 4\text{HWC}^2 + 2\text{HWM}^2C \\ &\quad + 2\text{NC}^2 + N^2C + NHWC.\end{aligned} \tag{12}$$

Comparing the complexity of GSSA in (12) with standard MSA, after subtracting the same term in both equations, we can obtain the ratio as shown in the following:

$$\frac{C_r(\text{GSSA})}{C_r(\text{MSA})} = \frac{M^2}{\text{HW}} + \frac{C}{M^2\text{HW}} + \frac{1}{2M^4} + \frac{1}{2M^2} \tag{13}$$

where $C_r(\text{GSSA}) = 2\text{HWM}^2C + 2\text{NC}^2 + N^2C + NHWC$ and $C_r(\text{MSA}) = 2(\text{HW})^2C$. The ratio in (13) is less than 1

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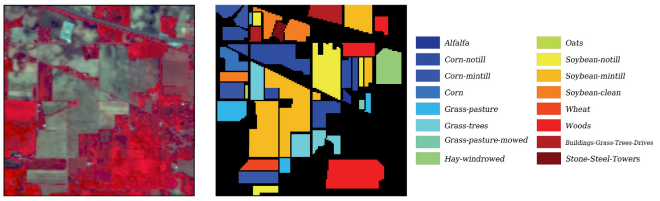
by adjusting the sizes of the relevant parameters. So, we can reduce the computational complexity of our GSSA relative to the standard MSA. In later experiments, we set $M \geq 4$ and $M \leq (1/2)H, M \leq (1/2)W$. Moreover, we can effortlessly keep $(C/M^2\text{HW}) \leq (1/16)$. Therefore, $(C_r(\text{GSSA})/C_r(\text{MSA})) \leq (1/4)+(1/16)+(1/512)+(1/32) = (177/512) < 1$. In a nutshell, GSSA enables the extraction of local and global spatial features, addressing the issue that the standard MSA has strong global dependency while lacks local dependency. Moreover, GSSA exhibits greater flexibility, allowing for the adjustment of patch sizes (H, W) and group spatial sizes M according to the specific requirements.

5) Enhancement of the MLP Layer: In a standard transformer block, the MLP layer is used to perform nonlinear transformation and mapping for features at each position, so as to extract higher-level and abstract feature representations. However, prior to the MLP layer, the model has already learned significant discriminant features for HSIs through GSC and GSSA. Therefore, performing complex nonlinear transformations is unnecessary. In later experiments, we show that one single PWC layer can substitute a typical MLP layer with two 2-D PWC operations without significantly affecting the classification performance of HSI. Additionally, after substituting the MLP layer with PWC, the number of convolution kernel parameters is $(1/2a)$ times the original parameters quantity, where a represents the expansion factor of MLP.

Also in a standard transformer block, as shown on the right of Fig. 2, both MSA and MLP layers are followed by a skip connection. This involves adding feature maps from the input MSA and MLP layers as a branch to the output of the MSA and MLP layers, so as to mitigate gradient vanishing and expedite convergence. Of course, this design is more pronounced when multiple transformer blocks are stacked, allowing the model to gain deeper semantic insights. However, considering the unique characteristics of HSIs, this architectural choice does not confer a significant advantage. Consequently, in our model, each FE Block contains only one single transformer block, and we directly connect the output of the *Conv block* with the output of the *PWC* in GS-ViT (as shown on the left of Fig. 2), thus facilitating multiscale features fusion of convolution-ViT.

III. Experimental Results and Discussions

This section is organized as follows. Section III-A describes four hyperspectral datasets used for assessing the GSC-ViT model. Section III-B conducts comparative analysis on different methods in SC Block. Section III-C conducts ablative experiments on the GSC block. Section III-D conducts ablative experiments on the GS-ViT block. Section III-E analyzes the model performance under different settings of key parameters, containing the patch size along with the group spatial size, the number of heads in GSSA, and the number of FE block in GSC-ViT. Section III-F compares the classification performance of our GSC-ViT model with some popular network models. Finally, Section III-G analyzes the computational complexity of the GSC-ViT model.



(a) (b)

Fig. 8. (a) False color composition of the AVIRIS Indian Pines scene. (b) Ground truth-map containing 16 mutually exclusive land-cover classes.

As a prelude, the parameter settings in our experiments are given first, where, the size of the input feature map is initialized to $C \times 8 \times 8$, where C represents the number of spectral bands. After passing through the SC block, the value of C is transformed to 256. Following the first FE block, the size of the feature map becomes $128 \times 8 \times 8$, and after the second FE block, it remains $64 \times 8 \times 8$.

For the training of the model, the input image patches are regularized to the range of [0, 1]. Besides, data augmentation techniques are applied to the input patches. To elaborate, each image patch is subjected to horizontal or vertical flipping, followed by a random rotation of 90°, 180°, or 270°. Moreover, *AdamW* optimizer is employed for parameter updating over a span of 200 training epochs. The learning rate is set to 0.001, and a weight decay coefficient of 0.05 is applied. The batch size is set to 128. The loss function of the model is the cross-entropy loss. To mitigate the impact of randomness and ensure the reliability of the results, we conduct each experiment ten times and record the average result.

The overall accuracy (OA), average accuracy (AA), and κ coefficient are employed to assess the performance of different classification methods. Finally, it is important to underline that all our experiments are conducted using Python 3.7 and Pytorch 1.11 on a desktop PC equipped with an NVIDIA GeForce RTX 2080 Ti and 32 GB of RAM.

A. Hyperspectral Datasets

In our experiments, four benchmark hyperspectral datasets are used to evaluate the performance of our proposed GSC-ViT model.

1) Indian Pines: The Indian Pines dataset was acquired by using the NASA AVIRIS instrument over the Indian Pines region in northwestern India in 1992. This dataset comprises 145×145 with a spectral range spanning from 0.4 to 2.5 μm. Although the ground objects were imaged across 220 consecutive spectral bands, 20 water-absorptive bands are typically omitted. The dataset encompasses 16 distinct and mutually exclusive ground-truth classes. Fig. 8(a) and (b) depicts false color composite of the image and corresponding ground-truth map. This dataset consists of 10 249 labeled samples and is split into 5% as the training set, 5% as the validation set, and 90% as the testing set, as detailed in Table I.

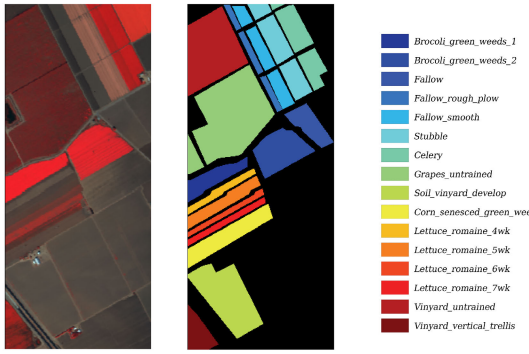
2) Salinas: The Salinas dataset was captured by using the AVIRIS imaging spectrometer in Salinas Valley, California, USA. It boasts a spatial resolution of 3.7 m, spanning a size of 512×217. Similar to the Indian Pines dataset, 20 water-absorptive wavebands were omitted, leaving a total of 204 remaining wavebands. Fig. 9 displays a false color

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TABLE I

NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH LAND COVER CLASS IN THE INDIAN PINES DATASET

Class No.	Land Cover Type	Train Num	Valid Num	Test Num
C1	Alfalfa	3	2	41
C2	Corn-notill	71	72	1285
C3	Corn-mintill	42	41	747
C4	Corn	12	12	213
C5	Grass-pasture	24	24	435
C6	Grass-trees	37	36	657
C7	Grass-pasture-mowed	2	1	25
C8	Hay-windrowed	24	24	430
C9	Oats	1	1	18
C10	Soybean-notill	49	48	875
C11	Soybean-mintill	122	123	2210
C12	Soybean-clean	30	29	534
C13	Wheat	10	10	185
C14	Woods	63	63	1139
C15	Buildings-Grass-Trees-Drives	19	20	347
C16	Stone-Steel-Towers	4	5	84
	Total	513	511	9225



(a) (b)

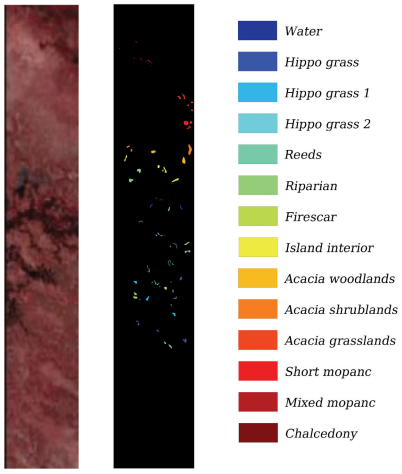
Fig. 9. (a) False color composition of the AVIRIS Salinas scene. (b) Ground-truth map containing 16 mutually exclusive land-cover classes.

TABLE II

NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH LAND COVER TYPE IN THE SALINAS DATASET

Class No.	Land Cover Type	Train Num	Valid Num	Test Num
C1	Brocoli-green-weeds-1	20	20	1969
C2	Brocoli-green-weeds-2	37	37	3652
C3	Fallow	20	20	1936
C4	Fallow-rough-plow	14	14	1366
C5	Fallow-smooth	27	27	2624
C6	Stubble	40	39	3880
C7	Celery	36	36	3507
C8	Grapes-untrained	113	112	11046
C9	Soil-vinyard-develop	62	62	6079
C10	Corn-senesced-green-weeds	33	33	3212
C11	Lettuce-romaine-4wk	11	10	1047
C12	Lettuce-romaine-5wk	20	19	1888
C13	Lettuce-romaine-6wk	9	9	898
C14	Lettuce-romaine-7wk	10	11	1049
C15	Vinyard-untrained	73	72	7123
C16	Vinyard-vertical-trellis	18	18	1771
	Total	543	539	53047

composite of the image alongside its corresponding ground truth map. In this dataset, a total of 54 129 samples from 16 distinct crop categories have been labeled, with specific numbers allocated for training (1%), validation (1%), and testing (98%) samples, as outlined in Table II.



(a) (b)

Fig. 10. (a) False color composition of the Botswana scene. (b) Ground-truth map containing 14 mutually exclusive land-cover classes.

TABLE III

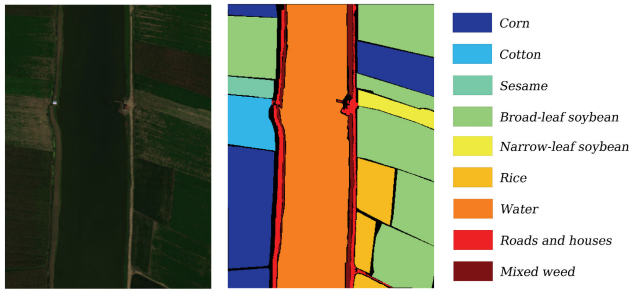
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH LAND COVER TYPE IN THE BOTSWANA DATASET

Class No.	Land Cover Type	Train Num	Valid Num	Test Num
C1	Water	14	13	243
C2	Hippo-grass	5	5	91
C3	Hippo grass-1	13	12	226
C4	Hippo grass-2	11	10	194
C5	Reeds	14	13	242
C6	Riparian	14	13	242
C7	Firescar	13	13	233
C8	Island interior	10	10	183
C9	Acacia woodlands	16	15	283
C10	Acacia shrublands	13	12	223
C11	Acacia grasslands	15	15	275
C12	Short mopanc	9	9	163
C13	Mixed mopanc	14	13	241
C14	Chalcedony	5	5	85
	Total	166	158	2924

3) Botswana: The Botswana dataset was captured by the NASA EO-1 satellite in the Okavango Delta region of Botswana in May 2001. The image has spatial dimensions of 1476 × 256, covering a wavelength range from 0.4–2.5 μm, and the spatial resolution is approximately 20 m. Out of the 242 spectral bands available, noise bands (1–9, 56–81, 98–101, 120–133, 165–186) were excluded, resulting in 145 bands for actual use. There are a total of 14 distinct land cover categories in the study area, representing land cover types of seasonal wetlands, intermittent wetlands, and dry woodlands located at the far end of the delta. The total number of samples is 3248. The training (5%), validation (5%), and testing (90%) divisions used in the experiments are listed in Table III.

4) WHU-Hi-LongKou: This dataset was collected in the town of LongKou, Hubei Province, China [40], [41]. It was acquired using an 8 mm focal length headwall hyperspectral imaging sensor mounted on the DJI Matrice 600 Pro unmanned aerial vehicle (UAV) platform. The data area represents a relatively straightforward agricultural scene, encompassing six crop types: corn, cotton, sesame, wide-leaf soybean, narrow-leaf soybean, and rice. The UAV flew at an altitude of 500 m, capturing images with dimensions

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(a) (b)

Fig. 11. (a) False color composition of the WHU-Hi-LongKou scene. (b) Ground-truth map containing nine mutually exclusive land-cover classes (right).

TABLE IV
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH LAND COVER TYPE IN THE WHU-HI-LONGKOU DATASET

Class No.	Land Cover Type	Train Num	Valid Num	Test Num
C1	Corn	69	69	34373
C2	Cotton	17	16	8341
C3	Sesame	6	6	3019
C4	Broad-leaf soybean	127	126	62959
C5	Narrow-leaf soybean	9	8	4134
C6	Rice	24	23	11807
C7	Water	134	134	66788
C8	Roads and houses	15	14	7095
C9	Mixed weed	11	10	5208
	Total	412	406	203724

of 550 × 400 pixels. The hyperspectral data spans from 0.4–1.0 μm, consisting of 270 spectral bands. The spatial resolution of the UAV-borne HSI is approximately 0.463 m. The total number of samples is 204 542. The training (0.2%), validation (0.2%), and testing (99.6%) divisions used in the experiments are listed in Table IV.

B. Analysis on Different Methods in SC Block

In this section, we make a comparative analysis of different spectral calibration methods in the SC block. Two alternative methods are compared with the PWC method we adopted: The first involves random channel replication/deletion (RCR/D), which is to randomly select a channel from the original channels of hyperspectral data and then replicated or deleted in situ, thereby ultimately calibrating the number of channels to a fixed value. The second method employs mirroring channel replication/deletion (MCR/D) on both ends, where only the channels at both ends are replicated or deleted to achieve a fixed number of channels. Both RCR/D and MCR/D employ nonparametric methods for the preprocessing of hyperspectral data, primarily focusing on enhancing local spectral features. RCR/D exhibits stochastic characteristics, while MCR/D is limited to operation at both ends. In addition, both ignore the global information of channels in the SC block.

The experimental results are shown in Table V, where two metrics are used: *Time* and *OA*. *Time* indicates the time to infer all test samples, and OA represents the overall accuracy of the model. It is evident from Table V, RCR/D and MCR/D exhibit significant disparities in OA results across

TABLE V
METRICS RESULTS OF RCR/D, MCR/D, AND PWC METHODS ON FOUR DATASETS

	Methods	Time/s	OA(%)
Indian Pines	RCR/D	3	95.04±0.64
	MCR/D	2	95.02±0.37
	PWC	**2**	**97.12±0.41**
Salinas	RCR/D	18	90.19±1.21
	MCR/D	13	91.16±0.88
	PWC	**11**	**97.15±0.47**
Botswana	RCR/D	50	84.20±1.11
	MCR/D	40	85.28±1.77
	PWC	**35**	**98.85±0.96**
WHU-Hi-LongKou	RCR/D	30	96.07±1.16
	MCR/D	28	97.21±0.31
	PWC	**27**	**98.65±0.20**

different datasets. When compared to our PWC method, the performance gaps are relatively smaller in the Indian Pines and WHU-Hi-LongKou datasets, while in the Salinas and Botswana datasets, the differences in OA results are pronounced. This discrepancy arises due to the distinct spectral characteristics of various hyperspectral datasets. Moreover, local spectral calibration methods tend to overlook spectral continuity and global features, posing a disadvantage for HSIC. Besides, both RCR/D and MCR/D necessitate separate channel-wise processing, resulting in longer training times than the PWC method.

C. Ablation Experiment in GSC Block

This section aims to the convolution operation in the GSC block, we compare and analyze the standard convolution and various GC strategies, including PWC combined with GC (PWC+GC) and GC only, as well as our proposed GSC. In addition, we also assess the impact of the number of groups in all compared GC strategies.

The experimental results are shown in Tables VI and VII, where four metrics are adopted: *Parameters*, *FLOPs*, *Time*, and OA. *Parameters* represents the number of parameters in the entire network with the selected convolution module. *FLOPs* signifies the number of floating-point operations performed by the model.

As observed from Tables VI and VII, when the Group Nums equals 1 ($g = 1$), it represents the standard convolution within the model. When we employ GSC instead of SC, it will bring a parameter reduction of approximately 290% on the Indian Pines dataset ($g = 8$), around 340% on the Salinas dataset ($g = 16$), roughly 330% on the Botswana dataset ($g = 8$), and about 270% on the WHU-Hi-Longkou dataset ($g = 16$). In addition, our GSC method exhibits advantages (such as smaller parameters and FLOPs) in comparison with the other two convolutional operations (PWC+GC and GC). In terms of overall performance, with the exception of the Botswana dataset where it is slightly lower than GC (but still fairly close), our GSC outperforms the other three convolution operations by a significant margin on the remaining datasets. However, in terms of inference time, GC is somewhat faster than GSC and PWC+GC, due to GSC and PWC+GC both involve two convolution operations while GC only involves

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TABLE VI
METRICS RESULTS OF DIFFERENT GROUP NUMBERS AND CONVOLUTIONAL OPERATIONS ON INDIAN PINES AND SALINAS DATASETS

Group Nums	Methods	Indian Pines				Salinas			
		Parameters/K	FLOPs/M	Time/s	OA(%)	Parameters/K	FLOPs/M	Time/s	OA(%)
1	SC	457.55	29.25	2	96.78±0.34	458.58	29.31	10	96.02±0.55
2	GSC	201.74	12.86	2	97.07±0.32	202.77	12.93	11	96.53±0.51
	PWC+GC	222.24	14.17	2	97.06±0.38	223.25	14.23	10	96.30±0.90
	GC	273.23	17.45	2	97.11±0.46	274.26	17.51	10	96.26±0.56
4	GSC	145.42	9.26	2	96.92±0.58	146.45	9.32	11	96.51±0.64
	PWC+GC	176.14	11.22	2	97.05±0.15	177.17	11.29	11	96.41±0.52
	GC	181.07	11.55	2	97.05±0.23	182.1	11.62	10	96.42±0.37
8	GSC	117.26	7.45	2	**97.12±0.41**	118.29	7.52	11	96.81±0.66
	PWC+GC	153.1	9.45	2	96.79±0.62	154.13	9.81	11	96.83±0.75
	GC	134.55	8.6	2	96.69±0.62	136.02	8.67	10	96.22±0.56
16	GSC	**103.18**	**6.55**	**2**	96.98±0.25	**104.21**	**6.6**	11	**97.15±0.47**
	PWC+GC	141.58	9.01	2	96.87±0.37	142.61	9.08	11	96.17±0.85
	GC	111.95	7.13	2	96.97±0.32	112.98	7.2	**10**	96.23±0.69

TABLE VII
METRICS RESULTS OF DIFFERENT GROUP NUMBERS AND CONVOLUTIONAL OPERATIONS ON BOTSWANA AND WHU-HI-LONGKOU DATASETS

Group Nums	Methods	Botswana				WHU-Hi-LongKou			
		Parameters/K	FLOPs/M	Time/s	OA(%)	Parameters/K	FLOPs/M	Time/s	OA(%)
1	SC	443.34	28.34	38	97.62±1.32	475.02	30.39	27	98.00±0.55
2	GSC	187.53	11.96	37	98.00±0.11	224.87	14.29	27	98.13±0.55
	PWC+GC	208.01	13.27	32	97.54±0.85	245.35	15.6	28	97.90±0.33
	GC	259.02	16.55	**32**	98.30±0.75	296.36	18.88	**26**	97.62±0.69
4	GSC	131.21	8.36	36	98.10±0.46	168.55	10.69	27	98.30±0.31
	PWC+GC	161.93	10.32	37	98.20±0.46	199.27	12.65	29	98.11±0.33
	GC	166.86	10.65	35	98.11±0.51	204.2	12.98	28	98.06±0.56
8	GSC	103.05	6.55	36	98.85±0.96	140.39	8.89	27	98.25±0.47
	PWC+GC	138.89	8.85	35	98.84±0.48	176.233	11.18	28	98.37±0.39
	GC	120.78	7.7	33	**98.86±0.65**	158.12	10.03	28	98.27±0.36
16	GSC	**88.97**	**5.65**	36	98.08±1.06	**126.31**	**7.98**	27	**98.65±0.20**
	PWC+GC	127.37	8.11	36	97.88±0.56	164.71	10.44	29	98.39±0.31
	GC	97.7	6.23	34	97.95±0.87	135.08	8.56	28	98.36±0.39

one. Nevertheless, the marginal sacrifice in time performance results in improved accuracy and reduced parameter quantity.

In terms of the number of groups, our GSC module exhibits a trend that the larger the group numbers, the lower the parameters and FLOPs, while the inference time changes little. However, the optimal group numbers for achieving the highest OA varies across different datasets. In the Indian Pines and Botswana datasets, the model achieves the highest OA when the group numbers are set to 8. While for the Salinas and WHU-Hi-LongKou datasets, the highest OA is obtained with 16 group numbers. We speculate that this variability is due to differences in the distributions of spectral band features across various datasets. Finally, to strike a balance between performance and the number of parameters, we set the group numbers to the value that yields the best results in each comparative experiment.

D. Ablation Experiment in GS-ViT Block

In this experiment, we conduct an ablative analysis on GS-ViT block. As shown in Table VIII, **MLP* method denotes the standard ViT block incorporating MLP layers, *MLP* signifies the utilization of MLP layers with conventional residual connection mechanisms within the GS-ViT, and *PWC* represents our proposed groupwise separable self-attention (GSSA), wherein the MLP layer is replaced with PWC, and accompanied by modifications to the residual connection scheme. The evaluation metrics include *Parameters*, *FLOPs*, *Time*, and OA. It should be noted that in our PWC scheme, we chose the best-performing group settings in *GSSA* block, which have been presented in Tables VI and VII.

TABLE VIII
COMPARING THE MLP LAYER WITH ORIGINAL RESIDUAL CONNECTION TO OUR PWC LAYER WITH A DIFFERENT RESIDUAL CONNECTION APPROACH ON FOUR DATASETS

	Methods	Params/K	FLOPs/M	Time/s	OA(%)
Indian Pines	*MLP	352.27	22.35	2	96.10±0.58
	MLP	214.54	13.63	2	96.82±0.65
	PWC	**117.26**	**7.45**	**2**	**97.12±0.41**
Salinas	*MLP	353.29	22.41	**10**	95.84±0.90
	MLP	215.57	13.7	12	96.28±0.32
	PWC	**104.21**	**6.6**	11	**97.15±0.47**
Botswana	*MLP	338.06	21.45	37	96.84±2.21
	MLP	200.33	12.7	41	98.25±0.61
	PWC	**103.05**	**6.55**	**35**	**98.85±0.96**
WHU-Hi -LongKou	*MLP	369.73	23.5	27	98.32±0.26
	MLP	237.67	15.06	29	98.11±0.45
	PWC	**126.31**	**7.98**	**27**	**98.65±0.20**

As shown in Table VIII, it is noteworthy that our model *PWC*, compared to **MLP*, exhibits an overall parameter reduction of approximately 200% on the Indian Pines dataset, around 239% on the Salinas dataset, roughly 228% on the Botswana dataset, and about 207% on the WHU-Hi-LongKou

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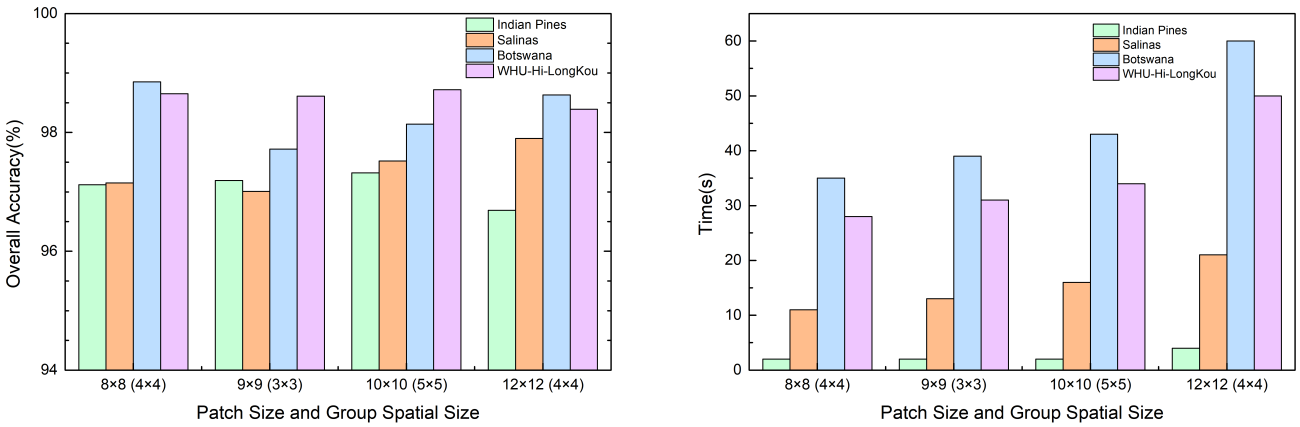


Fig. 12. OA and Times by using different patch sizes and group spatial sizes on four datasets.

TABLE IX

METRICS OF DIFFERENT NUMBERS OF FE BLOCKS AND HEADS ON INDIAN PINES AND SALINAS DATASETS

		Indian Pines				Salinas			
Block Nums	Head Nums	Parameters/K	FLOPs/M	Time/s	OA(%)	Parameters/K	FLOPs/M	Time/s	OA(%)
1	[1]	**104.02**	**6.52**	**1**	96.74±0.24	**93.77**	**5.87**	**9**	96.43±0.37
1	[2]	113.77	7.08	1	96.95±0.41	103.54	6.24	9	96.34±0.34
1	[4]	136.37	8.21	1	96.43±0.88	126.13	7.55	10	95.90±0.72
2	[1,1]	117.26	7.45	2	**97.12±0.41**	104.21	6.6	11	**97.15±0.47**
2	[2,2]	132.69	8.3	2	96.91±0.25	120.4	7.46	12	96.34±0.62
2	[4,4]	169.68	10.01	2	96.64±0.19	157.39	9.17	12	96.31±0.37
2	[8,8]	269.01	12.53	2	96.51±0.51	255.95	12.7	12	96.64±0.35
2	[16,16]	564.43	0.96	2	96.46±0.65	551.37	20.12	12	95.65±0.50
3	[1,1,1]	122.83	7.77	3	96.61±0.28	109.07	6.89	15	96.38±0.43
3	[2,2,2]	141.72	8.76	3	96.19±0.22	128.11	7.88	15	96.09±0.44
3	[4,4,4]	189.16	10.78	3	96.14±0.49	175.41	9.9	15	96.35±0.23

TABLE X

METRICS OF DIFFERENT NUMBERS OF FE BLOCKS AND HEADS ON BOTSWANA AND WHU-HI-LONGKOU DATASETS

		Botswana				WHU-Hi-LongKou			
Block Nums	Head Nums	Parameters/K	FLOPs/M	Time/s	OA(%)	Parameters/K	FLOPs/M	Time/s	OA(%)
1	[1]	**89.67**	**5.62**	**27**	98.39±0.68	**78.41**	**4.89**	**22**	98.41±0.26
1	[2]	99.43	6.18	30	98.20±1.31	88.17	5.46	25	98.41±0.25
1	[4]	122.03	7.31	31	97.68±1.19	110.76	6.59	26	98.35±0.38
2	[1,1]	103.05	6.55	35	**98.85±0.96**	126.31	7.98	27	**98.65±0.20**
2	[2,2]	119.24	7.4	38	98.82±0.91	136.84	8.54	28	98.36±0.31
2	[4,4]	156.24	9.11	40	98.21±0.49	173.83	10.25	29	98.18±0.42
2	[8,8]	254.79	12.6	42	97.89±0.23	272.39	13.77	31	98.23±0.74
2	[16,16]	550.22	20.06	43	97.88±0.99	567.81	21.21	32	97.87±0.46
3	[1,1,1]	108.68	6.87	52	98.21±0.97	125.73	7.97	39	98.34±0.33
3	[2,2,2]	127.73	6.87	54	97.47±0.64	144.77	8.96	40	98.28±0.35
3	[4,4,4]	175.02	9.88	54	97.63±1.05	192.07	10.98	40	98.27±0.36

dataset. On the other hand, it is evident that our *PWC* along with the new residual connection strategy yields advantages across all performance metrics comparing to the MLP-based GS-ViT architecture (*MLP*). Regarding *parameters* and *FLOPs*, the model with *PWC* significantly reduces, nearly halving them. In terms of inference time, *PWC* outperforms *MLP* slightly due to the latter's two-layer convolution compared to *PWC*'s one-layer operation. It is noteworthy that our newly residual connection enables the fusion of multiscale features from the convolution block and a single transformer block, resulting in improved OA results for all datasets.

E. Analysis on the Settings of Key Parameters

In this experiment, we analyze the effects of various configurations of key parameters. First, different patch sizes along with group spatial sizes are used to assess the model's classification performance on four datasets. Then, two other key parameters are assessed, including the number of heads and the number of FE blocks.

For the aspect of patch size and group spatial size, as shown in Fig. 12, the left plot represents accuracy, and the right plot represents inference time, both adopt the optimal group numbers in GSC. Four different patch sizes and their corresponding group spatial sizes are compared, including $8 \times 8(4 \times 4)$, $9 \times 9(3 \times 3)$, $10 \times 10(5 \times 5)$, and $12 \times 12(4 \times 4)$, where the values in parentheses denote the group spatial sizes. Here, the group spatial size is related to the patch size, with the former should be divisible by the latter.

As observed from the left plot in Fig. 12, OA does not increase as the patch size grows larger, instead, it exhibits

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TABLE XI

OAs[%], AAs[%], AND $\kappa \times 100$ STATISTIC FOR THE INDIAN PINES SCENE

Class	Convolution based Networks				Transformer based Networks					
	2DCNN	SPRN	3DCNN	HybridSN	SpectralFormer	SSFTT	GAHT	MorphFormer	CAEVT	GSC-ViT
1	70.24±10.16	53.17±17.55	58.54±13.54	75.61±17.86	21.95±4.63	77.07±22.70	29.27±13.71	74.63±19.37	76.10±23.97	**84.39±18.55**
2	84.03±3.67	84.89±5.32	79.61±3.77	91.69±2.72	72.62±2.31	90.40±2.47	93.01±2.99	92.48±1.63	91.69±1.69	**96.90±1.43**
3	92.32±2.19	93.31±3.82	76.49±8.24	90.31±2.41	66.88±7.94	89.42±1.81	94.83±2.94	94.83±3.48	95.72±0.77	**98.55±1.21**
4	84.69±3.29	86.95±2.81	51.74±11.20	87.23±8.60	53.99±5.20	90.80±6.70	**91.64±3.34**	90.89±5.21	91.36±7.09	88.45±7.18
5	92.87±3.18	90.39±4.58	88.18±4.17	90.48±3.59	78.53±3.96	91.36±3.73	90.76±2.59	92.14±2.44	91.36±2.21	**92.97±2.82**
6	97.47±1.04	98.14±0.91	98.23±0.67	98.96±0.53	93.88±2.62	98.60±0.92	**99.18±0.81**	98.96±0.72	98.93±0.54	99.09±0.92
7	**95.20±7.76**	53.60±25.75	77.60±26.12	84.00±13.39	57.60±31.76	83.20±19.66	69.60±16.12	68.80±16.08	81.60±21.41	92.00±14.09
8	95.26±4.99	97.63±3.01	98.09±2.17	98.33±1.63	95.63±2.76	98.47±1.29	99.72±0.23	98.37±0.76	97.81±1.62	**100.00±0.00**
9	43.33±11.33	30.00±10.30	41.11±16.33	45.56±20.61	22.22±16.10	66.67±12.67	17.78±10.77	65.56±15.48	63.33±9.69	**78.89±8.16**
10	88.75±2.15	89.17±2.36	79.77±2.88	91.06±3.48	74.33±2.71	88.32±5.10	93.51±2.69	94.13±2.34	87.79±3.26	**95.43±2.78**
11	93.15±1.81	93.30±0.86	89.40±2.51	94.15±2.68	83.91±2.85	94.91±1.06	95.56±1.07	96.78±0.51	95.34±1.70	**97.66±0.62**
12	81.95±3.57	76.55±12.77	71.42±9.59	85.88±6.62	44.19±6.13	91.09±6.74	91.42±5.18	88.05±2.82	90.86±5.27	**95.43±3.89**
13	99.03±1.05	98.49±1.43	99.35±0.53	99.24±0.81	97.84±1.37	99.24±0.55	99.35±0.63	99.57±0.40	98.92±0.76	**99.57±0.40**
14	97.51±1.40	97.05±0.95	97.44±2.15	97.88±0.90	94.19±1.83	97.82±1.08	98.33±1.09	99.02±0.44	98.89±0.82	**99.21±0.64**
15	86.51±5.40	85.65±3.78	86.28±5.83	88.76±2.31	73.43±6.14	86.86±5.79	89.63±5.77	90.72±3.75	86.57±5.13	**95.45±4.36**
16	**96.43±3.28**	86.67±9.39	88.57±3.58	95.71±2.08	93.33±7.21	96.19±3.48	84.52±9.64	90.95±8.96	90.71±5.75	95.48±2.65
OA(%)	91.19±0.82	90.84±1.28	85.95±2.24	93.10±0.58	78.84±1.21	93.15±0.50	94.42±0.55	94.96±0.24	93.93±0.02	**97.12±0.41**
AA(%)	87.42±1.07	82.18±3.74	80.11±3.38	88.43±2.15	70.28±2.90	90.03±0.99	83.63±1.74	89.74±2.25	89.81±1.58	**94.34±0.79**
$\kappa \times 100$	89.95±0.94	89.56±1.47	83.91±2.59	92.12±0.67	75.80±1.40	92.18±0.57	93.64±0.63	94.25±0.27	93.08±0.02	**96.67±0.40**

TABLE XII

OAs[%], AAs[%], AND $\kappa \times 100$ STATISTIC FOR THE SALINAS SCENE

Class	Convolution based Networks				Transformer based Networks					
	2DCNN	SPRN	3DCNN	HybridSN	SpectralFormer	SSFTT	GAHT	MorphFormer	CAEVT	GSC-ViT
1	54.69±44.79	96.84±1.47	95.75±5.06	99.87±0.26	94.87±1.28	98.94±1.67	99.74±0.30	98.19±2.24	**99.96±0.06**	99.66±0.25
2	96.23±6.23	99.54±0.07	99.88±0.07	99.92±0.10	98.67±0.85	99.86±0.22	**100.00±0.00**	99.97±0.04	99.52±0.92	99.85±0.11
3	70.32±22.31	93.47±4.05	89.58±2.21	95.06±3.56	93.14±5.03	94.21±7.92	**98.16±1.20**	95.85±1.05	95.79±2.88	96.21±5.96
4	98.62±0.95	96.02±2.48	98.51±0.93	**99.43±0.64**	92.61±2.76	99.34±0.59	98.92±1.28	98.96±0.78	96.71±4.15	99.05±0.86
5	76.99±29.21	94.57±2.48	95.40±2.10	**98.94±0.70**	88.35±5.74	98.06±2.51	98.32±1.94	98.13±1.66	97.39±1.90	98.70±1.46
6	99.08±1.32	99.42±0.71	99.69±0.42	99.96±0.05	99.44±0.74	**99.98±0.02**	99.97±0.04	99.51±0.48	99.18±1.65	99.93±0.13
7	94.03±7.14	98.99±0.81	99.70±0.12	99.83±0.16	98.34±0.45	99.86±0.10	99.99±0.01	99.90±0.13	99.53±0.65	**99.99±0.02**
8	82.82±8.60	88.58±1.85	83.82±2.28	90.69±2.18	83.85±3.23	88.87±3.5	**93.16±3.92**	91.32±2.50	89.72±2.73	92.55±3.31
9	96.71±3.80	97.13±1.71	97.45±1.37	99.12±1.64	98.10±1.14	99.00±1.12	**99.96±0.06**	99.60±0.41	98.23±2.40	99.84±0.18
10	95.73±0.57	92.43±1.05	92.80±1.46	95.62±1.79	87.68±6.32	96.18±2.80	97.02±2.03	95.31±3.46	94.89±2.97	**97.38±2.10**
11	64.74±31.13	93.12±3.93	73.26±8.11	97.48±2.00	76.16±4.56	95.36±5.15	**98.72±0.87**	96.87±3.49	94.54±5.44	96.96±2.26
12	94.22±4.36	99.36±0.51	98.59±1.19	99.84±0.19	95.71±1.29	99.72±0.15	99.76±0.24	**99.86±0.16**	99.11±1.23	98.54±1.14
13	99.47±0.54	99.76±0.30	99.00±1.68	97.02±4.76	94.37±6.84	98.26±2.72	98.55±1.78	99.00±1.45	98.49±1.70	**99.82±0.26**
14	92.32±7.09	97.67±1.67	97.98±0.64	98.28±0.83	96.93±1.60	95.77±3.14	**99.24±0.21**	98.59±0.72	98.93±1.05	98.86±0.98
15	76.72±15.80	85.94±3.88	74.31±5.49	82.98±5.02	76.46±6.63	85.66±6.97	90.93±3.99	92.19±2.45	86.87±4.87	**92.74±3.02**
16	78.22±26.28	88.74±2.25	91.90±1.52	95.98±1.93	91.53±4.29	97.30±0.98	97.01±2.83	96.79±1.74	96.47±2.10	**97.78±1.70**
OA(%)	86.21±5.05	93.49±0.64	90.69±0.75	94.86±0.60	90.00±1.36	94.72±0.50	96.81±0.34	96.21±0.45	94.79±0.37	**97.15±0.47**
AA(%)	85.68±7.11	95.10±0.78	92.98±0.49	96.88±0.57	91.64±1.20	96.65±0.62	**98.09±0.21**	97.50±0.22	96.58±0.56	97.99±0.62
$\kappa \times 100$	84.63±5.67	92.76±0.71	89.64±0.83	94.28±0.67	88.87±1.51	94.13±0.56	96.45±0.38	95.79±0.50	94.20±0.41	**96.47±0.83**

variations across different datasets. Among, Indian Pines and WHU-Hi-LongKou achieve the highest OA with a $10 \times 10(5 \times 5)$ setting, while Salinas performs best with the largest patch size of $12 \times 12(4 \times 4)$. Botswana, on the other hand, shows superior performance with the smallest setting of $8 \times 8(4 \times 4)$. We speculate that these variations are attributed to the distinct spatial feature distributions of different HSI datasets, causing the optimal receptive field of our GSC-ViT to differ across datasets. Furthermore, in the right plot of Fig. 12, as the patch size increases, so does the inference time for all datasets. This is because the enlargement of the patch size leads to a decrease in the processing speed of all GSC and GSSA blocks. In fact, patch size does not have a significant impact on OA, and the variation in OA is minimal. Therefore, to ensure a lightweight and fast model, we select the smallest patch size and its corresponding group spatial size, that is $8 \times 8(4 \times 4)$.

For the aspect of the number of heads in GSSA and the number of FE blocks, as demonstrated in Tables IX and X, we configure the number of FE blocks as 1, 2, and 3, respectively. And with a block count of 1 and 3, the number of heads in GSSA is set to 1, 2, and 4 for each FE block. Meanwhile, with a block count of 2, we further increase the number of heads to 8 and 16 for each FE block. The evaluation metrics include *Parameters*, *FLOPs*, *Time*, and OA.

In the context of FE blocks, we can observe that with an equal number of heads, an increased number of FE blocks results in elevated model parameters, FLOPs, and inference time. For all datasets, when the block number is 2, OA is generally higher compared to block numbers of 1 and 3. Simultaneously, the highest OA is attained when the number of blocks is set to 2 and the number of heads is set to 1. In the context of heads in GSSA, it is evident that with the same block number, an increased number of heads leads to augmentation in model parameters, FLOPs, and inference time among all datasets. However, OA reaches its peak only when the number of heads is set to 1. It should be noted that the number of FE blocks affects the depth of feature extraction. Given the characteristics of HSI data, opting for an excessively deep network model is unnecessary, as it would result in a larger model scale and slower inference. Simultaneously, an increased number of heads, while causing an escalation in model parameters, might enable the model to extract more

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TABLE XIII

OAs[%], AAs[%], AND $\kappa \times 100$ STATISTIC FOR THE BOTSWANA SCENE

Class	Convolution based Networks				Transformer based Networks					
	2DCNN	SPRN	3DCNN	HybridSN	SpectralFormer	SSFTT	GAHT	MorphFormer	CAEVT	GSC-ViT
1	98.27±2.15	99.09±1.23	**100.00±0.00**	99.42±0.81	99.84±0.20	99.59±0.52	97.86±2.68	99.01±1.06	99.92±0.16	99.67±0.40
2	80.22±11.38	97.80±4.40	82.64±5.91	97.80±1.97	93.63±6.64	98.24±1.79	**100.00±0.00**	98.68±2.64	98.68±2.64	99.34±1.32
3	87.96±3.51	97.08±2.35	96.99±1.54	94.96±3.34	88.41±5.45	99.47±0.86	**100.00±0.00**	98.50±2.40	100.00±0.00	99.56±0.88
4	81.34±9.07	98.97±1.08	95.36±2.14	97.63±3.41	61.24±18.88	100.00±0.00	99.79±0.41	**100.00±0.00**	99.38±1.24	99.48±0.46
5	84.21±8.81	87.11±7.59	85.87±7.22	86.03±10.71	35.54±7.12	87.27±10.44	92.48±6.48	**95.87±3.95**	91.90±5.39	94.55±5.92
6	80.08±6.03	93.97±3.12	78.84±10.92	92.07±7.71	71.16±10.16	89.09±4.59	**97.93±1.20**	94.88±3.02	96.86±2.30	97.11±2.65
7	95.97±3.99	99.91±0.17	99.66±0.32	99.66±0.50	97.94±0.83	99.31±1.37	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
8	75.85±14.02	96.17±5.07	90.82±6.98	91.91±11.25	72.24±10.14	97.92±3.61	99.89±0.22	**100.00±0.00**	99.02±1.31	99.56±0.87
9	82.12±12.50	94.13±3.52	94.49±3.89	95.12±3.12	60.57±15.57	91.02±15.67	**98.66±1.99**	92.58±9.51	93.78±5.64	98.59±1.86
10	97.22±2.20	99.46±0.72	97.58±1.92	99.01±1.97	96.50±3.82	99.19±0.72	100.00±0.00	**100.00±0.00**	96.95±2.68	99.37±0.78
11	95.27±1.10	97.67±1.67	96.95±2.18	97.24±2.31	97.38±0.74	96.22±3.60	98.47±1.11	97.82±1.85	**99.71±0.58**	99.35±0.63
12	94.60±2.24	96.20±2.03	98.04±1.79	98.04±1.84	95.83±1.19	99.51±0.72	99.14±1.43	98.65±0.25	99.39±0.78	**99.88±0.25**
13	96.02±3.39	100.00±0.00	98.01±2.84	99.42±0.50	96.10±3.38	99.42±1.16	100.00±0.00	**100.00±0.00**	99.75±0.50	99.75±0.33
14	95.53±3.44	96.24±4.55	89.41±6.40	95.53±5.07	82.82±0.94	**99.53±0.58**	94.12±6.82	96.00±4.80	98.35±2.05	98.35±1.60
OA(%)	89.14±2.15	96.60±1.04	93.81±1.40	95.90±2.89	81.31±1.75	96.35±1.75	98.52±0.76	97.88±0.96	97.95±0.66	**98.85±0.96**
AA(%)	88.90±1.93	96.70±1.20	93.19±1.53	95.99±2.88	82.08±1.43	96.84±2.95	98.45±0.92	98.00±0.90	98.12±0.62	**98.90±0.54**
$\kappa \times 100$	88.23±2.33	96.32±1.13	93.29±1.51	95.55±3.13	79.76±1.88	96.05±1.89	98.39±0.82	97.70±1.04	97.78±0.72	**98.75±0.52**

TABLE XIV

OAs[%], AAs[%], AND $\kappa \times 100$ STATISTIC FOR THE WHU-HI-LONGKOU SCENE

Class	Convolution based Networks				Transformer based Networks					
	2DCNN	SPRN	3DCNN	HybridSN	SpectralFormer	SSFTT	GAHT	MorphFormer	CAEVT	GSC-ViT
1	96.70±4.00	97.84±1.52	98.03±0.72	99.55±0.31	99.01±0.52	99.50±0.34	99.68±0.16	99.46±0.51	99.50±0.27	**99.78±0.14**
2	50.72±42.09	85.20±4.75	77.46±3.84	95.15±2.41	81.62±9.33	96.09±2.18	96.87±1.08	97.16±1.42	94.13±4.09	**97.90±0.91**
3	0.00±0.00	47.19±33.08	54.78±13.54	78.90±7.32	**92.89±2.72**	85.27±7.26	86.88±7.95	89.20±3.61	88.26±6.22	90.50±4.93
4	95.84±1.81	97.85±0.61	97.37±0.60	97.59±0.71	96.99±1.05	99.08±0.38	99.13±0.47	**99.24±0.37**	98.44±0.53	98.94±0.51
5	15.78±31.55	20.68±6.82	41.49±13.28	79.64±2.59	66.47±9.33	83.72±3.70	85.22±4.50	82.01±4.38	82.16±4.89	**85.88±5.60**
6	66.69±22.52	91.84±7.15	93.97±4.65	96.26±2.07	88.08±2.13	96.76±2.56	**98.73±0.68**	97.71±1.64	98.24±1.62	97.14±2.42
7	99.98±0.02	99.95±0.07	99.98±0.02	99.92±0.14	99.90±0.16	99.90±0.14	99.84±0.19	**99.93±0.09**	99.82±0.25	99.89±0.18
8	78.74±27.83	85.40±6.39	87.79±1.36	88.99±2.01	75.09±5.85	93.14±2.45	92.55±2.55	92.60±1.73	91.81±4.09	**96.05±1.88**
9	39.36±19.96	56.89±10.97	68.61±10.47	80.16±5.38	44.60±7.80	83.53±6.87	81.04±7.60	76.72±7.48	**87.30±7.17**	83.73±9.40
OA(%)	88.72±3.91	93.87±1.21	94.49±0.51	97.12±0.26	94.36±0.61	98.04±0.23	98.18±0.31	98.02±0.33	97.89±0.17	**98.65±0.20**
AA(%)	60.42±8.51	75.87±5.31	79.94±2.07	90.68±0.83	82.74±1.89	93.00±1.21	93.33±1.94	92.67±1.31	93.30±1.05	**94.42±1.71**
$\kappa \times 100$	84.96±5.29	91.89±1.63	92.72±0.68	96.21±0.34	92.57±0.80	97.42±0.30	97.61±0.41	97.40±0.44	97.22±0.23	**97.85±0.52**



Fig. 13. Classification maps obtained by different models for the Indian Pines dataset. These maps correspond to a single experiment in Table XI.

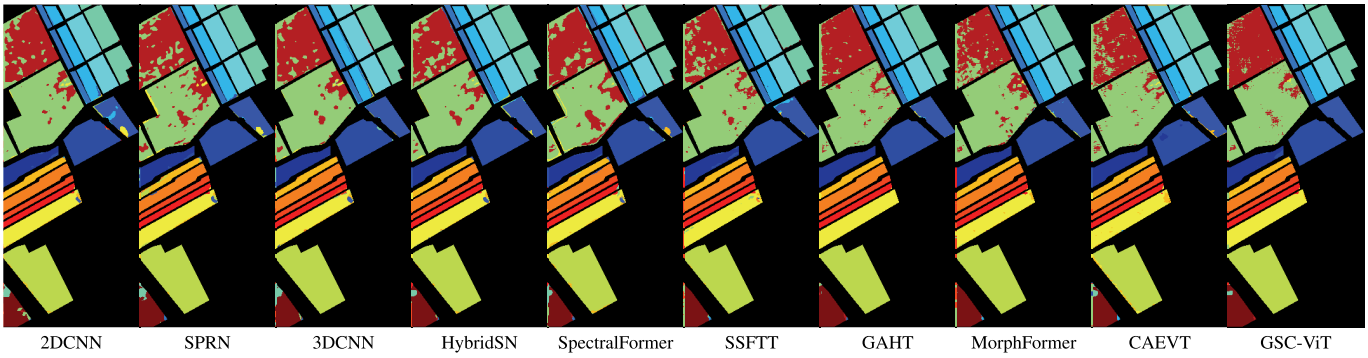


Fig. 14. Classification maps obtained by different models for the Salinas dataset. These maps correspond to a single experiment in Table XII.

intricate and redundant features of HSI. Comprehensively considering model performance and model scale, we set the number of FE blocks in our model to 2 and the number of heads in GSSA to 1.

F. Classification Comparison With State-of-the-Art Models

To further underscore the advantages of our proposed GSC-ViT, we conduct a comparative evaluation against the state-of-the-art methods covering the CNN and Transformer architectures. Specific descriptions are as follows.

1) *2-DCNN* [20]: This model comprises two 2-D CNN layers followed by two FC layers. After each CNN layer, a max-pooling layer is applied, and the final classification output is obtained through the FC layers.

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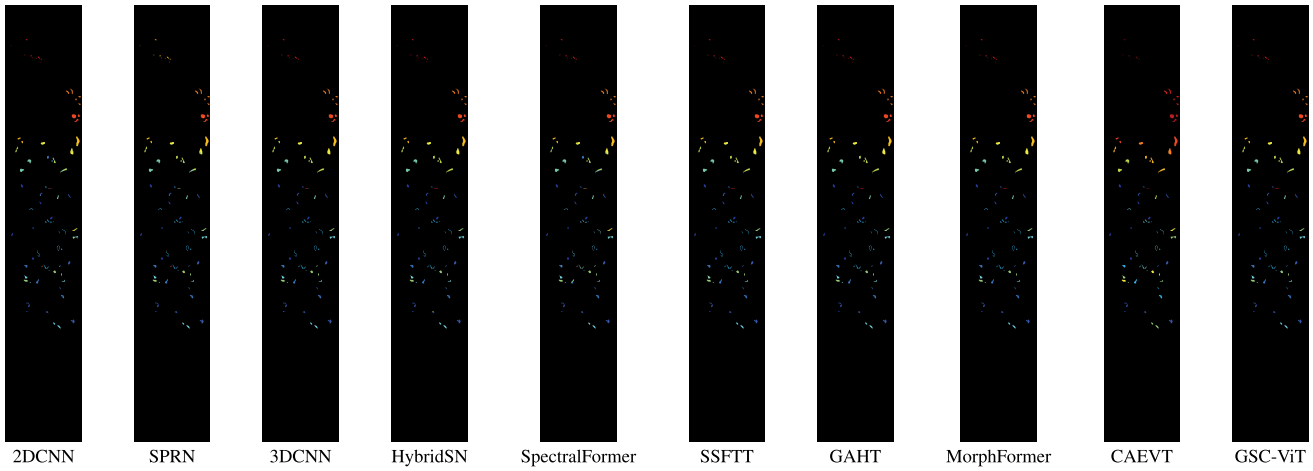


Fig. 15. Classification maps obtained by different models for the Botswana dataset. These maps correspond to a single experiment in Table XIII.

2) *SPRN* [42]: This model primarily consists of a padding layer, an optional spatial attention layer, two convolutional layers, two residual blocks, a FC layer, and a linear classification layer.
3) *3-DCNN* [21]: This model consists of three 3-D convolutional layers, two 3-D pooling layers, a dropout layer, and a FC layer. The first two 3-D convolutional layers are followed by two 3-D pooling layers, and finally, a FC layer is used for output classification.
4) *HybridSN* [22]: This is a mixed 2-D and 3-D CNN model, including three 3-D convolutional layers, one 2-D convolutional layer and three FC layers.
5) *SpectralFormer* [36]: This model includes a layer for processing spectral information in the neighbor band, a patch embedding layer, a transformer layer, and an MLP layer for classification.
6) *SSFTT* [37]: This model includes a 3-D convolutional layer, a 2-D convolutional layer, a transformer layer that incorporates positional encoding, a dropout layer, and a linear layer for classification.
7) *GAHT* [38]: This model consists of three convolutional layers, three transformer layers, a GAP layer, and a linear layer for classification.
8) *MorphFormer* [39]: This model consists of a convolutional layer, a grouped separable convolutional layer, a self-attention module based on morphology joint in spectral–spatial domain, and a linear layer for classification.
9) *CAEVT* [43]: This model initially employs a 3-D convolutional autoencoder (3D-CAE) constructed with a 3D-CNN to extract spatial–spectral features from the feature map. Subsequently, the feature map is fed into a lightweight MobileViT, and finally, a linear layer is utilized for classification.

Tables XI–XIV record the OA, AA, κ, and accuracies of each class for all compared methods on four HSI datasets. The best results are highlighted in bold. Obviously, our proposed GSC-ViT performs the best among all methods, with the highest OA, AA, and κ (except for AA in Salinas).

TABLE XV
PERFORMANCE COMPARISON OF TEN MODELS ON FOUR HSI DATASETS

	Model	Parameters/K	FLOPs/M	Time/s
Indian Pines	2DCNN	1718.6	67.56	2
	SPRN	183.35	11.81	2
	3DCNN	161.68	81.05	7
	HybridSN	512.22	247.68	4
	SpectralFormer	345.53	107.95	5
	SSFTT	931.85	49.38	**1**
	GAHT	830.8	159.35	2
	MorphFormer	199.6	19.46	4
	CAEVT	359.95	120.96	3
	GSC-ViT	**103.18**	**6.55**	2
Salinas	2DCNN	1718.16	67.56	11
	SPRN	183.35	11.81	12
	3DCNN	263.6	138.25	8
	HybridSN	512.22	247.68	18
	SpectralFormer	355.29	111.04	26
	SSFTT	950.28	50.37	**6**
	GAHT	972.62	62.18	11
	MorphFormer	202.8	19.84	24
	CAEVT	359.95	123.02	20
	GSC-ViT	**104.21**	**6.6**	11
Botswana	2DCNN	1581.97	51.11	52
	SPRN	183.35	11.81	48
	3DCNN	179.59	98.37	**19**
	HybridSN	512.22	247.68	49
	SpectralFormer	230.72	69.15	81
	SSFTT	678.34	35.74	20
	GAHT	800.72	51.2	33
	MorphFormer	194.93	16.78	71
	CAEVT	262.67	83.51	47
	GSC-ViT	**103.05**	**8.36**	36
WHU-Hi-LongKou	2DCNN	1869.32	85.96	32
	SPRN	183.35	11.81	32
	3DCNN	201.99	182.71	21
	HybridSN	512.22	247.68	36
	SpectralFormer	543.52	167.4	72
	SSFTT	1253.95	66.74	**17**
	GAHT	1514.2	96.85	29
	MorphFormer	255.14	26.09	52
	CAEVT	454.92	166.4	54
	GSC-ViT	**126.31**	**7.98**	27

In the Indian Pines dataset, we achieved an OA of $97.12 \pm 0.41\%$ using only 5% of the training samples, which is about 2% higher than the second place. The classification accuracy for the Hay-windrowed category reached 100%. In the Salinas dataset, we achieved an OA of $97.15 \pm 0.47\%$ using only

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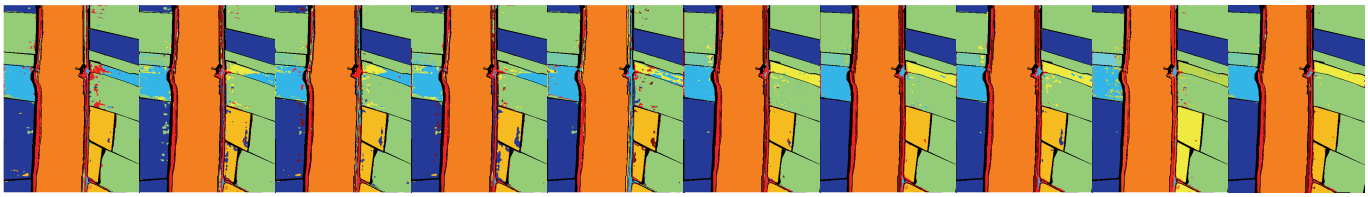


Fig. 16. Classification maps obtained by different models for the WHU-Hi-LongKou dataset. These maps correspond to a single experiment in Table XIV.

1% of the training samples, which is about 0.3% higher than the second place. Although our method may not outperform GAHT in terms of classification accuracy for some specific class, our OA is higher than GAHT, indicating that our model has good generalization performance. For the Botswana dataset, similar to Indian Pines, 5% of the samples were used for training, and our model achieved an accuracy of $98.85 \pm 0.96\%$, which is about 0.3% higher than the second-place. In the WHU-Hi-LongKou dataset, we set the training samples to 0.2% for the experiment. Despite such a small amount of training samples, our model still achieved an accuracy of $98.65 \pm 0.2\%$, which is about 0.5% higher than the second place. With ten repeated experiments, the OA fluctuation of our model was only 0.2%, indicating that our model is relatively stable on this dataset.

In addition, to specifically demonstrate the classification accuracy of each compared model, we present the classification maps of different models with their highest OA values of ten experiment in Figs. 13–16. Observed from these figures, our GSC-ViT classification map has less noise than other methods on the four datasets, and is closer to the ground-truth distribution.

G. Analysis on Computational Complexity

Certainly, the advantage of our GSC-ViT is not limited to accuracy. We have listed our model's *Parameters*, *FLOPs*, and *Inference Time* in Table XV compared to other models on four datasets. In terms of *Parameters* and *FLOPs*, our model is significantly smaller than other models, which is the main motivation for designing it, namely to achieve high classification accuracies with minimum parameters quantity. However, in terms of *Inference Time*, our model is not as fast as the SSFTT. This is because the SSFTT is shallower with fewer convolution and self-attention operations, while our GSC-ViT has more GC and self-attention operations, which slightly slows down the speed. Nevertheless, the *Inference Time* of our model is still competitive as compared to other models except for SSFTT. Its ranking keeps high among all compared models on the four datasets.

IV. Conclusion and Future Research

In this article, we developed a new lightweight network model, called GSC-ViT, for advanced land cover classification of remotely sensed HSIs. Our model is composed of a spectral calibration block, two feature extraction blocks, a GAP block and a FC layer. For each feature extraction block, we used GSC and GS-ViT separately. GSC includes GPWC and GC, which can reduce network parameters and effectively extract local spectral-spatial features. GS-ViT used GSSA instead of traditional MSA to reduce computation and extract local-global spatial features. In addition, we substitute the MLP layer with PWC in GS-ViT and improve the residual connection structure to effectively integrate the features of convolution-ViT to enhance the classification performance. Experiments compared different group schemes and selected the best settings for each dataset, and also compared different patch sizes and group spatial sizes, as well as the MLP layer with original connection with our PWC layer with an improved residual connection strategy. Finally, we compared our model with state-of-the-art classification models on four HSI datasets. From the comprehensively experimental results, we can see that our model can maintain excellent classification accuracy while reducing a significant number of parameters.

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